

Research

The impact of dam management and rainfall patterns on flooding in the Niger Delta: using Sentinel-1 SAR data

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Abstract

This study explored how dam operation and rainfall patterns interact to influence flooding in the Niger Delta, Nigeria. This study utilizes Sentinel-1 Synthetic Aperture Radar (SAR) data to analyze the flood events from 2018 to 2022, focusing on key locations such as Obiafu (downstream), Kainji/Jebba, and Lagdo dams (upstream). Analyzed Sentinel-1 satellite data [European Space Agency (ESA), 2018–2022], shuttle Radar Topography Mission, alongside rainfall data [Center for Hydrometeorology and Remote Sensing (CHRS), 2018–2022] and surface water level to assess flood variations between 2018 and 2022. The flood extent was differentiated from that of permanent water bodies using Sentinel-1 images with an RGB band (Snap 7.0), and the rainfall patterns were examined via inverse distance weighting (IDW) interpolation (ArcGIS 10.5). The findings reveal that the intensity and timing of flooding vary significantly across these regions. Obiafu experienced high flooding in October 2020 and 2022, while Kainji/Jebba and Lagdo dams had contrasting flood severities during these periods. Additionally, a comparative analysis of hydrograph data from these locations shows a delayed downstream response, highlighting the need for coordinated water release strategies to mitigate flood risks. The results also indicate that upstream dam operations, particularly water releases, contribute to downstream flooding, although local rainfall patterns remain a crucial factor. By aligning SAR data with real-time water levels, this study emphasizes the importance of integrating satellite-based monitoring with hydrological models to improve flood prediction accuracy and minimize socio-economic impacts in the Niger Delta. The study concludes that enhanced flood management strategies, incorporating both rainfall forecasting and dam operation schedules, are critical for reducing flood-related vulnerabilities in the region.

Highlights

- Sentinel-1 SAR effectively mapped flooding, with peak levels in October 2022 in the Niger Delta.
- September and October floods link rainfall peaks and dam releases, heightening flood risks.
- Coordinated dam water release could reduce downstream flood severity in low-lying areas.

Keywords Flooding · Dam operations · Sentinel-1 satellite data · Niger Delta · Rainfall patterns

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1 Introduction

The Niger Delta, a vast wetland region in southern Nigeria, is highly susceptible to a range of environmental challenges, with flooding being one of the most severe [1–3]. Flooding in this region poses significant threats to human life, infrastructure, and economic activities [4, 5]. Over the years, floods have caused massive damage, leading to long-term economic impacts. Like the devastating floods of 2012 and 2018 inflicted substantial economic losses, amounting to billions of naira, while destroying farmlands and essential infrastructure [5–9]. These recurrent flood events severely disrupt local economies and necessitate extended recovery periods, leaving affected communities struggling to rebuild. Mfon et al. [10] and Do et al. [11] emphasize the profound economic consequences of such floods, noting that they not only affect local livelihoods but also disrupt broader economic systems reliant on agriculture, commerce, and infrastructure. While natural rainfall patterns are an important factor in the Niger Delta's frequent flooding, there is increasing evidence that human-induced activities significantly exacerbate flood risks [10, 12–14]. Among these activities, the construction and operation of dams and reservoirs along the Niger River and its tributaries have profoundly altered the region's natural water flow [15–17]. This disruption has contributed to more intense and frequent flooding events downstream, especially in the Niger Delta [8]. The construction of major dams, such as the Kainji, Jebba, and Lagdo dams, as well as smaller reservoirs, has disrupted the region's hydrological balance [18–20]. These dams, designed primarily for hydropower and irrigation, have significantly altered the seasonal flow regime, resulting in flood episodes during peak rainy seasons in downstream areas like the Niger Delta [14, 19, 21]. Adding to the region's vulnerability are other human-related activities such as deforestation, urbanization, and poor environmental management. According to Alborzi et al. [22] and Fagunloye [23], anthropogenic influences, including poorly planned urban growth and deforestation, amplify the natural effects of rainfall, leading to more frequent and severe flooding events. Reports from the Nigeria Hydrological Services Agency (NIHSA) and the United Nations Office for Disaster Risk Reduction (UNISDR) have consistently warned about rising water levels during the rainy season in the Niger Delta. These institutions stress the need for comprehensive flood management strategies to mitigate flood-related risks in the region [24, 25]. However, the translation of these warnings into concrete action remains limited, as inadequate flood control infrastructure and suboptimal flood mitigation measures continue to persist in the region. The operation of dams, although beneficial in generating electricity and providing water for irrigation, has a profound impact on river flow regimes [26, 27]. The interruption of natural water flows can exacerbate downstream flooding, particularly during heavy rainfall events [28, 29]. Dams often alter the volume and timing of water releases, which can overwhelm downstream floodplains, resulting in severe flood events [30–32]. Additionally, dams trap sediment that would naturally replenish downstream ecosystems, causing riverbeds to erode and increasing flood risks [33, 34]. The dual impacts of altered hydrology and sediment transport have created serious ecological and socioeconomic consequences in the Niger Delta, a problem highlighted by Dada et al. [35], who emphasize the urgent need for sustainable flood management interventions. Understanding the complex interplay between dam management practices and rainfall patterns is critical for mitigating flood risks in the Niger Delta [19]. This study aims to explore the influence of dam operations, specifically in conjunction with regional rainfall trends, on downstream flood events. By utilizing Sentinel-1 synthetic aperture radar (SAR) imagery, the study seeks to provide a detailed analysis of flood dynamics in the Niger Delta. Sentinel-1 SAR imagery offers a distinct advantage for flood monitoring, as it can penetrate cloud cover and gather data regardless of weather conditions, making it highly reliable for real-time flood mapping in tropical regions like the Niger Delta [36–38]. Sentinel-1, a mission under the European Space Agency's Copernicus program, has emerged as a valuable tool for Earth observation, particularly in flood-prone areas [39, 40]. Unlike optical sensors, which are often hindered by cloud cover, SAR technology operates in all weather conditions, making it ideal for flood monitoring in regions like the Niger Delta where cloud cover is persistent during rainy seasons [39, 41, 42]. The ability of SAR to capture high-resolution data during periods of heavy rainfall is crucial for accurate flood mapping and management, as optical data acquisition is often not feasible during such times [40, 43–45]. This study leverages the capabilities of Sentinel-1 SAR to monitor flood events, providing critical insights into the spatial and temporal patterns of flooding in the Niger Delta. This research contributes to the growing body of literature on the application of Earth observation techniques for flood management in complex river systems. By focusing on the interactions between natural rainfall patterns and human-induced alterations (such as dam operations), this study provides actionable insights that can assist government agencies, environmental organizations, and local communities in developing more effective flood management strategies. The findings are expected to enhance flood preparedness and response efforts in the Niger Delta, helping to protect lives, infrastructure, and the broader ecological integrity of the region. The study adopts a hydrological framework to analyze the dynamics of flood events in the Niger Delta, focusing on the impact of dam operations and rainfall patterns.

Hydrology, as a field, provides a comprehensive understanding of the movement, distribution, and management of water resources, making it a suitable theoretical lens for this research [46, 47]. This framework allows for the exploration of the interactions between rainfall, dam operations, and flood risks, particularly within the context of riverine systems like the Niger Delta [19]. The hydrological cycle and its components, such as precipitation, runoff, and water storage, are central to understanding the causes and consequences of flooding in the region [48].

The research also incorporates disaster studies theory, which explores the social and environmental factors that contribute to disaster risks, such as flooding. By integrating disaster studies into the theoretical framework, the study emphasizes the impact of human activities, including dam construction and land-use changes, on increasing flood vulnerability [49, 50]. This interdisciplinary perspective enhances the analysis by addressing both the physical and social aspects of flood risks in the Niger Delta [51]. Nonetheless, there are gaps in the existing literature, particularly concerning the integration of dam management data with real-time flood monitoring through advanced satellite technology. While prior studies have examined the individual effects of rainfall and dam operations on flooding, few have combined these elements to offer a comprehensive analysis of flood dynamics in the Niger Delta. This research seeks to bridge that gap by merging dam operation data, rainfall trends, and SAR-derived flood maps to develop a more holistic understanding of the drivers of flood events in the region.

This study aims to investigate how dam operations, in conjunction with rainfall patterns, influence flood occurrences in the Niger Delta. The primary research question guiding the study is: How do dam operations and rainfall patterns interact to influence the frequency and severity of flooding in the Niger Delta? To address this question, the study has four specific objectives: (1) to analyze the historical and seasonal patterns of rainfall in the Niger Delta and their contribution to flooding; (2) to assess the role of upstream dam operations, such as those of Kainji, Jebba, and Lagdo dams, in influencing downstream flood patterns in the region; (3) to evaluate the effectiveness of Sentinel-1 SAR data in mapping and monitoring flood events in the Niger Delta; and (4) to provide recommendations for improving flood mitigation strategies, with a focus on dam management and real-time flood monitoring in the Niger Delta.

2 Methodology

2.1 Study area

The Niger Delta, which is situated in southern Nigeria, encompasses a vast and intricate network of rivers, wetlands, and mangrove forests. Spanning more than 70,000 square kilometers [52], this region plays a vital role in Nigeria's ecology and economy. The specific extent of the Niger Delta can vary slightly depending on the source. However, it generally covers parts of six states: The Edo, Delta, Rivers, Bayelsa, Akwa Ibom, and Cross River [53]. between N 4° 52' 30" and N 5° 03' 00" in latitude and between E 6° 12' 00" and E 6° 25' 30" in longitude [53] (Fig. 1). The geomorphology of the Niger Delta is characterized by low-lying plains with a network of small rivers and streams that drain into the Atlantic Ocean. This intricate system is prone to flooding due to the region's low elevation and high water table [7, 41]. Mangrove forests and freshwater marshes are the dominant vegetation types and provide vital habitat for a diverse range of flora and fauna. However, this rich ecosystem faces significant challenges. The inherent susceptibility to flooding was highlighted by Amangabara and Obenade [54]. Moreover, the application of open-access remotely sensed data for flood modeling and mapping in developing regions is a major concern [41]. Additionally, human activities such as oil exploration and production have had detrimental impacts on the environment [55].

Figure 1 shows the DEM of the study area, which captures the varying elevations across the Niger Delta and its neighboring regions, including the locations of major dams such as Kainji, Jebba, and Lagdo. The DEM is instrumental in understanding the flood dynamics, as elevation plays a crucial role in water flow and accumulation [56]. The study area's elevation ranges from 78 to 1399.43 m, as shown in the elevation bands, which correspond to different dam locations: Obiafu (Niger Delta) is situated at a lower elevation of 78–235.1 m. Kainji/Jebba and Lagdo dams are located at higher elevations, with the Kainji/Jebba dam ranging from 235.11 to 342.73 m and the Lagdo dam extending up to 1399.43 m. Elevation plays a critical role in determining the extent and severity of flooding, particularly in regions with extensive river networks and low-lying deltaic systems, such as the Niger Delta. The varying elevation ranges outlined in the DEM provide insights into the area's most likely to experience significant water accumulation during heavy rainfall and dam releases [56] Low-lying areas, particularly in the Niger Delta, are prone to flooding due to their limited capacity for water drainage, resulting in water logging and extensive flood plains. Higher-elevation areas such as those around Kainji/Jebba

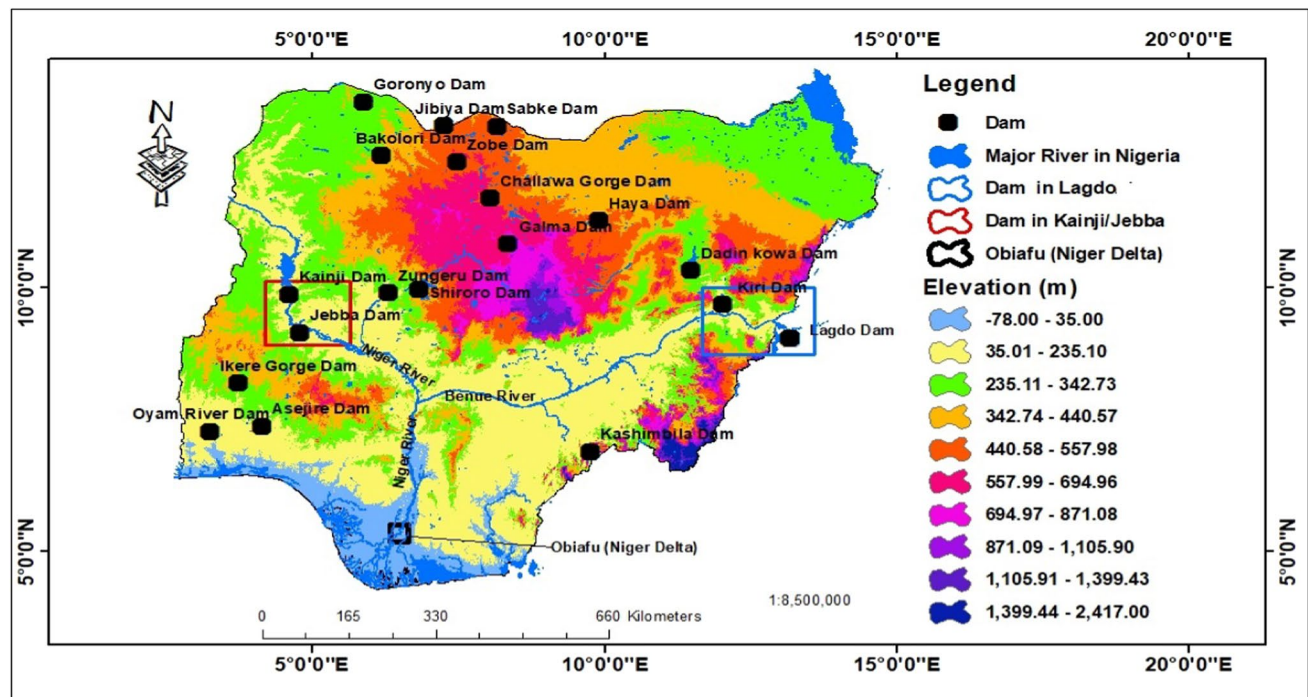


Fig. 1 Digital elevation model of Study area showing dams in Nigeria

and Lagdo dams have a lesser risk of direct flooding but can influence water discharge levels downstream, increasing flood risks in the Niger Delta region.

2.2 Role of upstream reservoirs in downstream flood control

Upstream reservoirs play a pivotal role in regulating downstream water levels, particularly in regions like the Niger Delta where flooding can have severe socio-economic and environmental impacts [15]. The presence of major reservoirs, such as the Kainji, Jebba, and Lagdo dams, enables the management of water flow and flood control by storing water during high inflow periods and releasing it gradually during drier seasons. This control is essential in mitigating downstream flood risks, especially during periods of heavy rainfall or extreme weather events [25, 57]. In the study area, the Jebba and Lagdo dams, located upstream, are critical in influencing water levels in downstream areas such as Obiafu in the Niger Delta. These reservoirs, constructed primarily for hydropower generation, have significant storage capacities, which allow them to regulate the release of water, thus reducing the peak flow of floodwaters downstream. However, the timing and volume of water releases from these reservoirs must be carefully managed to prevent exacerbating flood risks downstream [20].

The reservoirs within the study area, particularly Jebba and Lagdo, have different storage capacities and operational procedures, which influence their role in flood control. For example, Jebba Dam, located on the Niger River, plays a crucial role in regulating the flow of water toward the Niger Delta. Similarly, the Lagdo Dam, situated upstream in Cameroon, impacts water flow into the Benue River, contributing to downstream flood conditions. The coordinated operation of these dams, especially during the rainy season when inflows are at their peak, is vital to flood mitigation strategies in the region [20]. Upstream reservoirs have been recognized for their significant role in regulating water flow, mitigating flood risks, and controlling downstream water levels, especially in flood-prone areas such as the Niger Delta. These reservoirs, when strategically managed, act as buffers, capturing excessive water during periods of heavy rainfall and gradually releasing it to prevent downstream flooding [58, 59]. In the Niger Delta region, dam operations from upstream reservoirs such as Kainji, Jebba, and Lagdo, are key in determining flood outcomes downstream. It is essential to provide detailed information about these reservoirs to understand their hydrological influence. Reservoirs, like those mentioned above, are designed to store large volumes of water and can influence water availability, drought control, and flood risk management in downstream areas. The timing, capacity, and operational rules of these reservoirs play a critical role in

flood control. Like the release of stored water can either prevent or exacerbate downstream flooding depending on how it aligns with rainfall patterns [60].

Kainji Reservoir, situated on the Niger River, was constructed primarily for hydroelectric power generation but also plays a crucial role in flood control. Its reservoir has a capacity of approximately 15 billion cubic meters, providing significant storage space to control water flow downstream during periods of excessive rainfall [57, 61, 62]. During peak flood events, the Kainji reservoir has the potential to reduce flood peaks downstream by releasing water in a controlled manner [63]. Jebba Reservoir, located downstream of Kainji, acts as a secondary flood buffer. Together with Kainji, these reservoirs form a critical water management system that regulates water flow along the Niger River. The operational coordination between Kainji and Jebba is vital for managing flood risks, particularly in the Niger Delta region, where high rainfall often coincides with reservoir releases [61, 64]. The Lagdo Dam, situated on the Benue River in Cameroon, affects flood patterns in the Niger Delta, particularly when its waters are released simultaneously with heavy rainfall downstream [65]. The Lagdo reservoir can store approximately 7.7 billion cubic meters of water, and its operation requires coordination with downstream flood control mechanisms to mitigate flood impacts on communities in the Niger Delta [66].

2.3 Materials and methods

A complex network of wetlands and rivers in the Niger Delta faces increasing challenges from flooding. Dam construction activities along upstream tributaries are suspected to alter natural flow patterns and potentially exacerbate these events. This study explored the potential impact of dam activities on flooding in the Niger Delta using Sentinel-1 satellite data, a valuable tool for Earth observation due to its all-weather imaging capabilities [67]. The methodology focuses on data acquisition, preprocessing, and processing techniques to generate flood maps and monitor flood events over a 5-year period (August to October, 2018–2023).

The analysis focused on mapping and monitoring flooded areas and permanent water bodies over a 5-year period, from 2018 to 2022. Synthetic aperture radar (SAR) is a powerful energetic remote sensing technology that is used in a variety of applications, including flood monitoring. Radar is an active system that illuminates the Earth's surface, so images can be collected during the day under any lighting conditions or at night in the complete absence of daylight. Furthermore, these images are unaffected by bad weather, fog, or smoke. As a result, radar signals can pass through these barriers [68]. Due to their low or no roughness in the absence of waves, continental waters behave like a mirror surface, reflecting the radar signal in the opposite direction of the sensor position; thus, continental waters exhibit very low backscatter values (negative) in comparison to those of other land covers. Because of the high contrast in backscatter values, SAR images can be used to precisely map flooded areas.

Synthetic aperture radar (SAR) data are of special interest for observing flood situations and can be used for the following purposes: flood mapping, flood detection, global flood monitoring system (GFMS), and flood prediction.

2.4 The SAR method

- The RGB composite method

The RGB composite method is used to perform multitemporal modifications and to detect changes in the terrain surface using a temporal color image [69]. It can also distinguish between permanent water bodies and flooded areas; as a result, a multitemporal image is created in which a band is assigned to each of the primary colors to form an RGB composite image [70].

2.5 Data acquisition and processing

2.5.1 Sentinel-1 data and software

The Sentinel-1 mission of the European Space Agency (ESA) provides radar imagery essential for Earth observation applications (<https://www.esa.int/>) and provides high-resolution radar imagery suitable for flood monitoring applications [71]. The Sentinel Application Platform (SNAP) software, version 7.5, serves as the primary tool for preprocessing and analyzing the Sentinel-1 data used in this study [SNAP. 2023] [72].

2.6 Preprocessing steps

The raw Sentinel-1 data undergo several preprocessing steps, as shown in Fig. 2, to enhance the quality of the data and prepare the data for subsequent analysis [73].

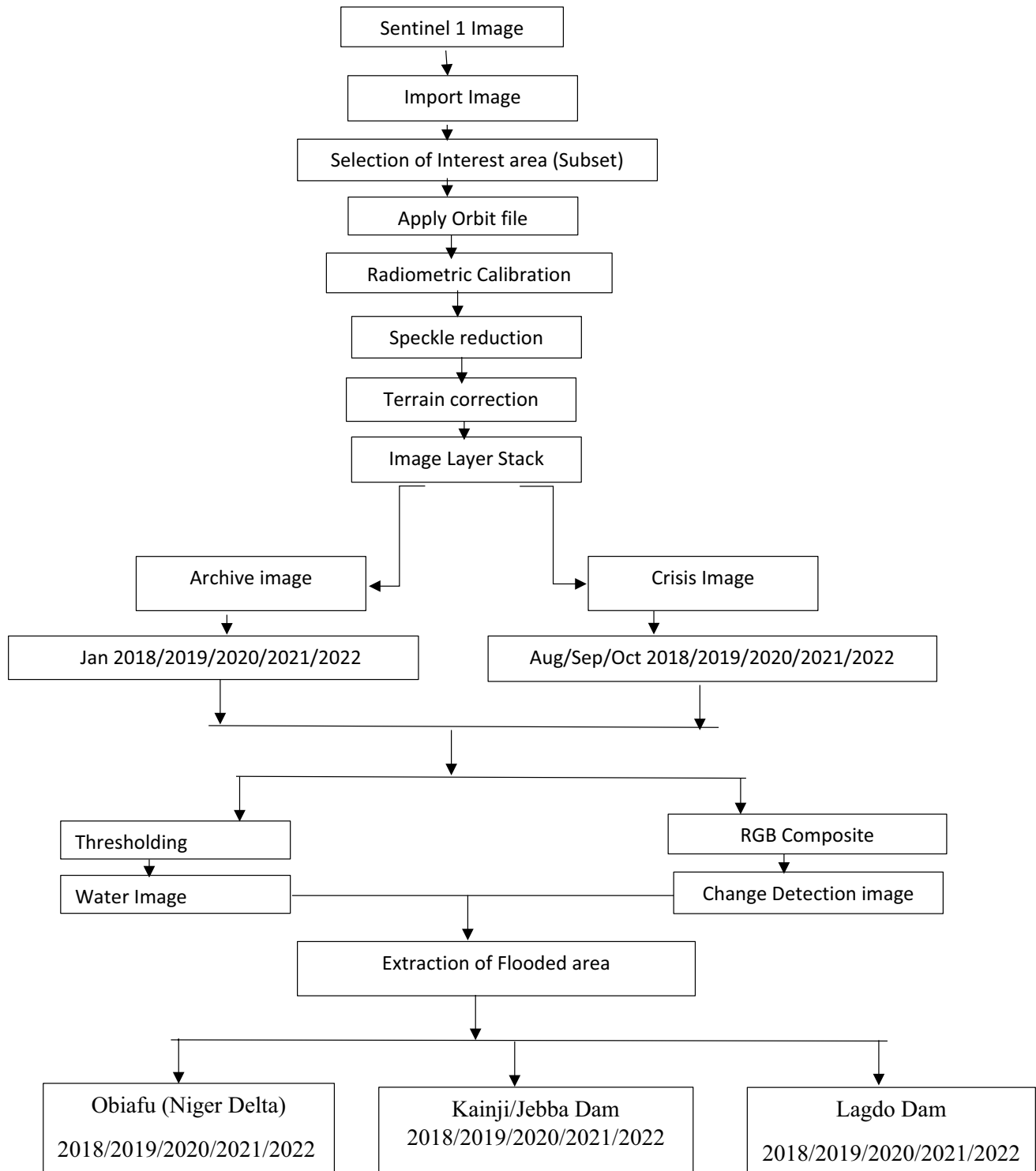


Fig. 2 Workflow chart for preprocessing used for Sentinel-1 images

Radiometric calibration: This step converts the raw digital numbers (DN) stored in the satellite imagery into meaningful physical units—radar backscatter coefficients (σ^0) [74, 75]. These coefficients represent the amount of radar energy reflected back to the satellite sensor by the Earth's surface, providing valuable information about surface properties such as the presence of water [76].

Speckle filtering: Speckle noise is an inherent characteristic of radar imagery and appears as a grainy texture that can hinder data interpretation [77]. The Lee filter is applied in this stage to suppress speckle noise while preserving image details critical for flood analysis [75, 78].

Geometric correction: To ensure accurate spatial representation and enable precise location referencing within the Niger Delta, the imagery undergoes georeferencing. This process involves employing a digital elevation model (DEM) to account for distortions caused by the satellite's viewing geometry and Earth's curvature [79].

2.7 Rainfall data

To complement the satellite imagery and establish correlations between potential flood events and rainfall patterns, daily precipitation data from the Center for Hydrometeorology and Remote Sensing (CHRS) covering the period March to September of each year from 2018 to 2023 were utilized [80]. This timeframe encompasses the rainy season in the Niger Delta, allowing for a comprehensive analysis of rainfall patterns preceding and coinciding with flood occurrences [81].

2.8 Software and processing methodology

ArcGIS software version 10.5 was used as the primary platform for rainfall data analysis and visualization. The processing steps involve the following steps:

Data formatting: The CHRS rainfall data were downloaded and converted into a format compatible with ArcGIS to facilitate seamless integration and analysis.

Spatial interpolation: The inverse distance weighting (IDW) interpolation technique was used to generate spatial rainfall maps across the Niger Delta region. The Inverse Distance Weighting (IDW) interpolation technique, its effectiveness in representing continuous data in regions with limited measurement points, like the Niger Delta. IDW assumes that the closer the points, the more similar they are, assigning weights based on distance. This method provides a simple and accurate representation of rainfall data across unsampled areas, ensuring smooth transitions and maintaining high spatial resolution [82]. Unlike methods like kriging, which may require more computational effort and model assumptions, IDW is straightforward and offers a balance between accuracy and simplicity, making it ideal for rainfall interpolation in this specific geographic context.

Digital Elevation Model (DEM): The Shuttle Radar Topography Mission (SRTM) DEM was used to assess the topography of both upstream and downstream areas. DEM data helped to analyze how elevation influences the extent of flooding. The DEM was processed in Google Earth Engine to delineate flood-prone areas based on elevation, slope, and water flow direction. Data were exported to ArcGIS 10.5 for further visualization.

Surface water level: Surface water levels for Kainji/Jebba and Lagdo dams were collected from the Database for Hydrological Time Series of Inland Waters (DAHITI), developed by the Deutsches Geodätisches Forschungsinstitut at Technische Universität München. DAHITI provides high-resolution satellite altimetry data essential for monitoring water bodies. Additionally, data from the Nigeria Hydrological Services Agency (NIHSA) were utilized to complement the hydrological information. These datasets enabled accurate analysis of water level fluctuations at both upstream and downstream reservoirs, which are critical for understanding the influence of dam operations on flooding in the Niger Delta. In addition, ground truth data were collected through field surveys using measuring tools to acquire surface water level in study area at Obiafu from 2018 to 2022 and was analyze using python program to plot the line chart.

2.9 Limitations

Rainfall data resolution: Although IDW interpolation works well with sparse data, it may not capture localized rainfall extremes due to the limited number of gauging stations, leading to potential underestimations or overestimations in certain areas.

SAR data limitations: Sentinel-1 SAR data, while robust for flood detection, may face challenges in detecting floods under dense vegetation cover or mixed land uses, which can distort backscatter values.

DEM accuracy: The accuracy of flood extent mapping depends heavily on the DEM quality. Low-resolution DEMs may lead to misinterpretations of flood-prone areas, particularly in flat regions where small elevation changes affect flood behavior.

Temporal resolution: The study's reliance on Sentinel-1 SAR data collected every 12 days may miss sudden, short-duration flood events, limiting the capture of peak flood conditions.

2.10 Flood mapping and monitoring

Following preprocessing, the backscatter coefficient (σ_0) from the Sentinel-1 data was utilized for flood mapping. The RGB composite is employed to identify flooded areas based on its distinct backscatter properties compared to those of dry land and permanent water bodies [11]. The generated flood maps are correlated with the processed rainfall data to identify potential linkages between dam activity and flood events. By analyzing a time series of flood maps (2018–2023), this study aimed to monitor the spatial and temporal extent of flooding in the Niger Delta. This information can be used to assess potential changes in flood patterns that might be attributed to dam operation.

3 Results and discussion

3.1 Results

See Tables 2, 3, 4; Figs. 3, 4, 5, 6, 7.

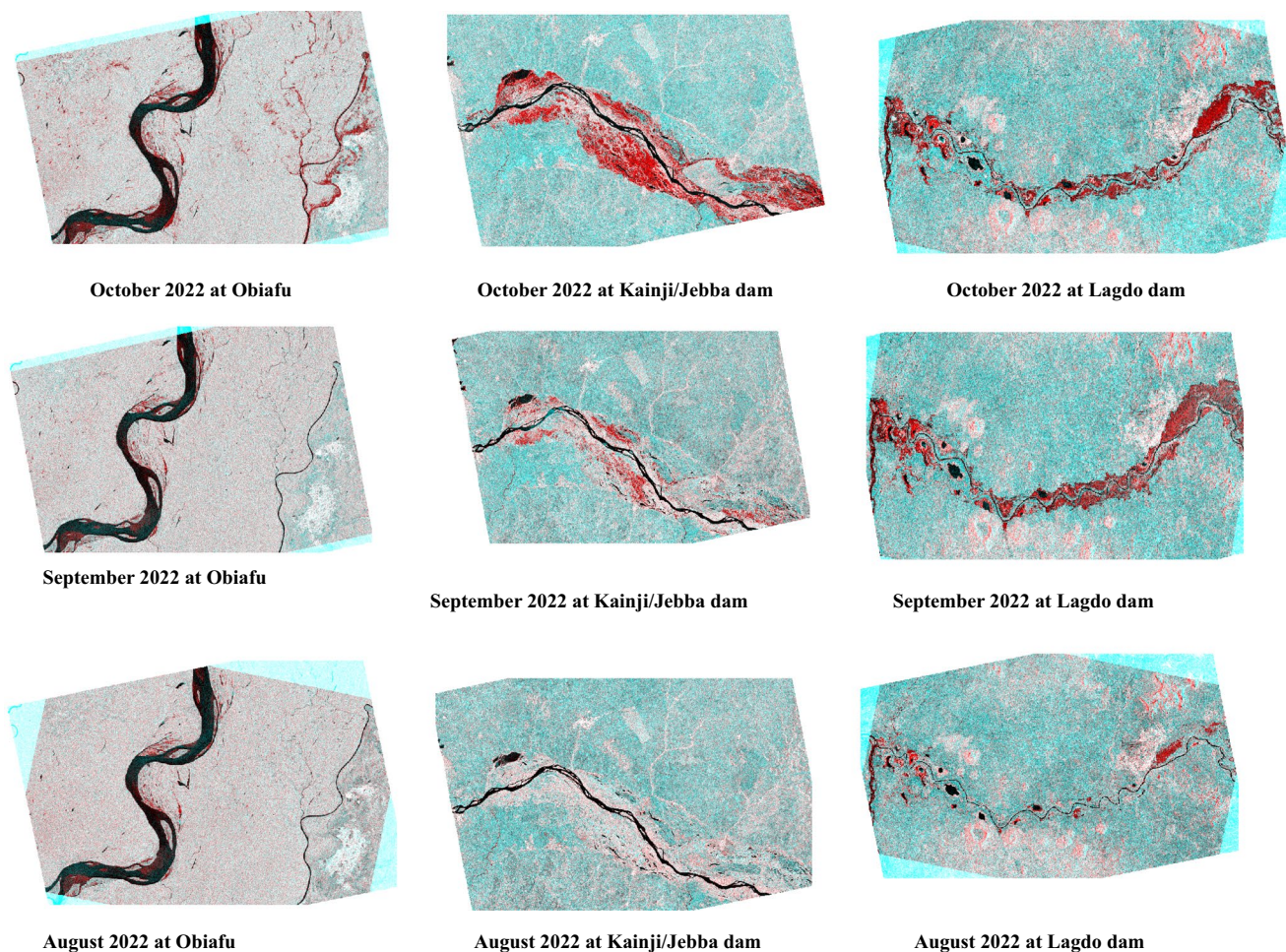


Fig. 3 SAR model result from August 2022 to October 2022

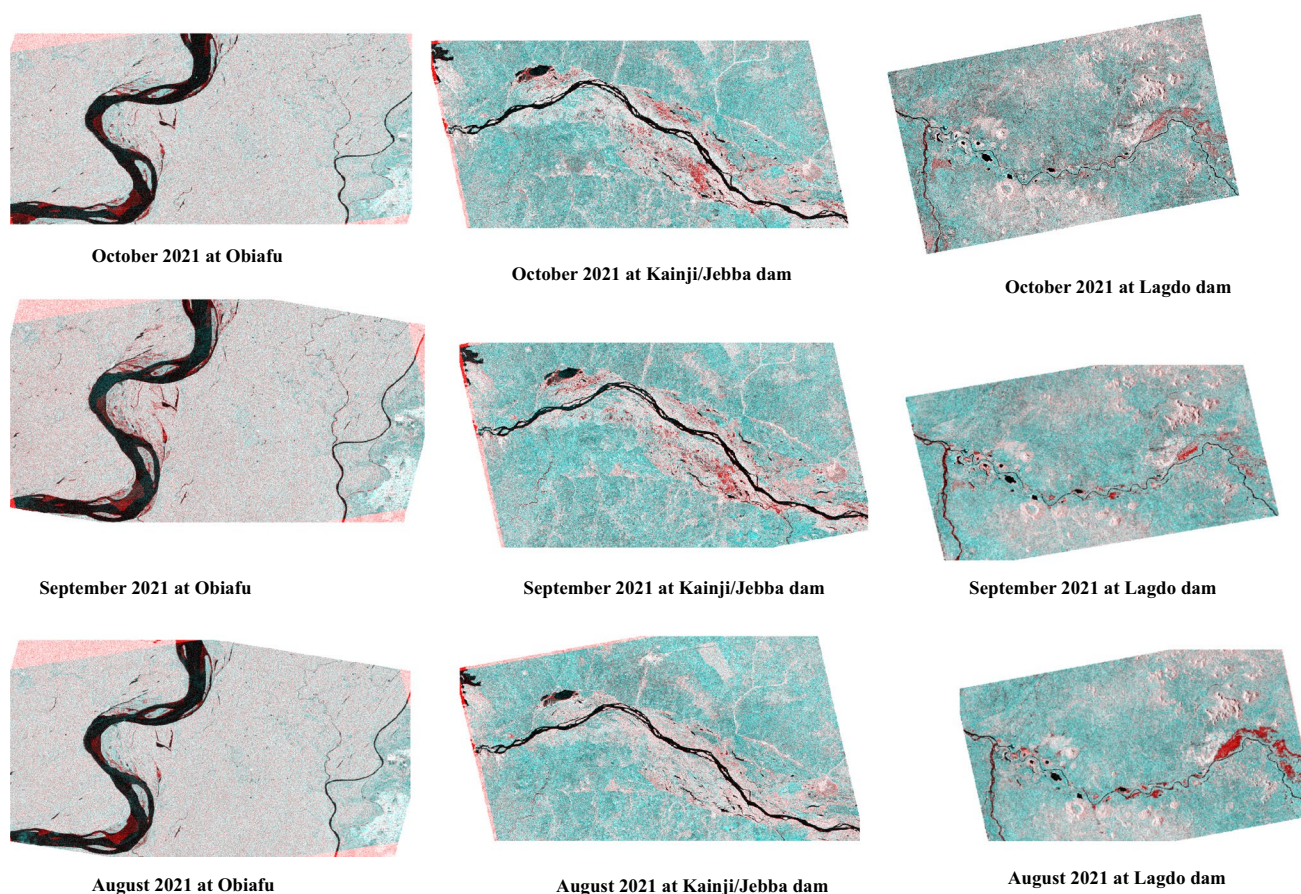


Fig. 4 SAR model result from August 2021 to October 2021

4 Discussion

4.1 Flood mapping with Sentinel-1 SAR

We employed Sentinel-1 SAR data and the RGB band technique to differentiate flooded areas from permanent water bodies in Obiafu. The data were acquired using Snap 7.0 software for both the preflood period (January) and during flood events (August, September, and October) across the 5 years. The red band in the RGB image represents flooded areas with a high radar signal response, while black indicates permanent water bodies with a low response. The green and blue bands represent areas with low responses, indicating that they were not flooded (Figs. 3, 4, 5, 6, and 7).

4.2 Flood ranking and analysis

These findings are in line with the flood ranking levels in Table 1. Tables 2, 3, 4 present the flood rankings derived from the SAR model for the Obiafu community, Kainji/Jebba dam, and Lagdo dam. The flood rankings were categorized based on the severity of flooding observed in each month over the 5-year period. These findings are in line with the flood ranking levels in Table 1. Tables 2, 3, 4 present the flood rankings derived from the SAR model for the Obiafu community, Kainji/Jebba dam, and Lagdo dam. The flood rankings were categorized based on the severity of flooding observed in each month over the 5-year period.

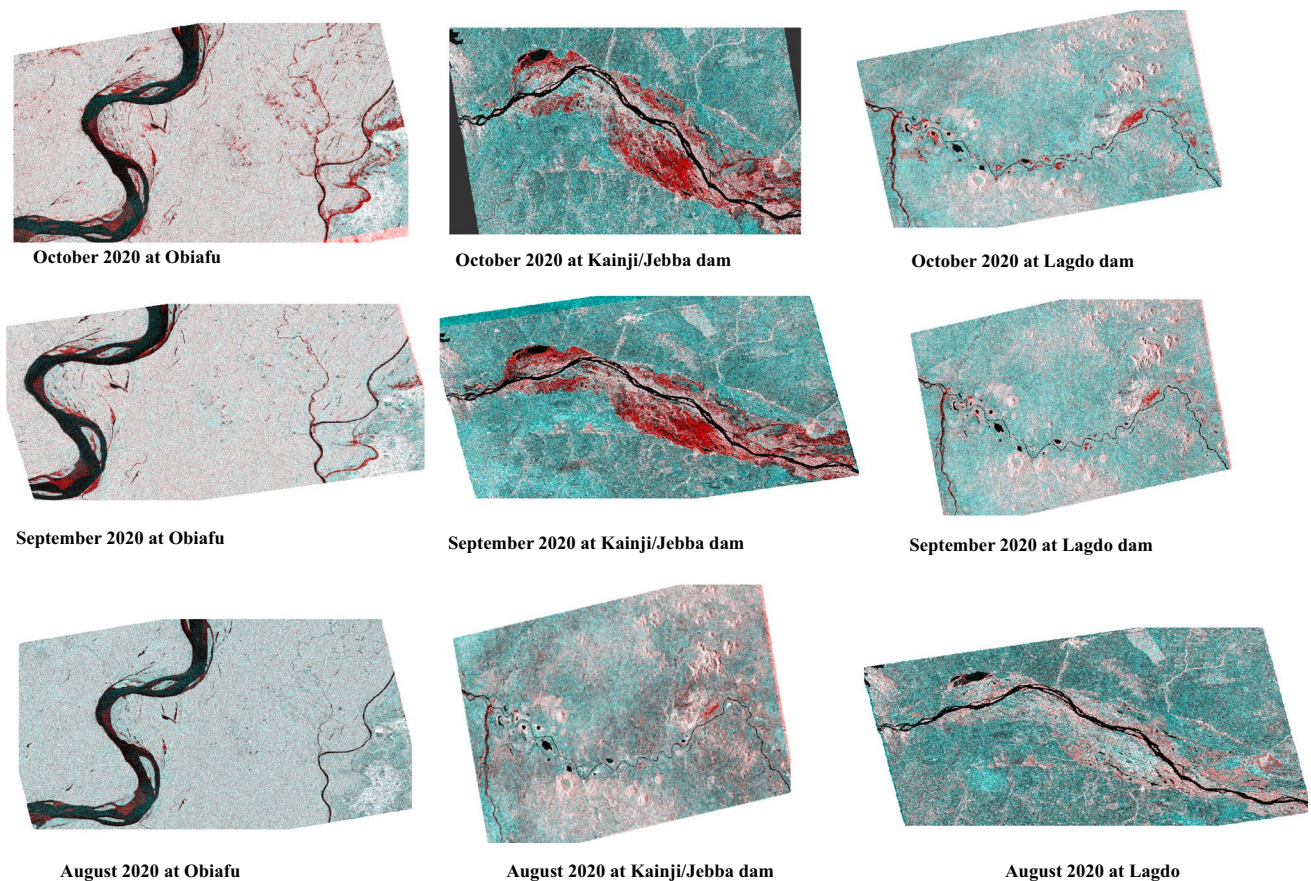


Fig. 5 SAR model result from August 2020 to October 2020

4.3 October flooding analysis (Table 2)

In October 2021, the SAR model indicated very low flooding in the Obiafu community, Kainji/Jebba, and Lagdo dams. Conversely, in October 2022, the flooding rate was very high at all three locations. In October 2020, Obiafu and Kainji/Jebba experienced very high flooding, while Lagdo experienced low flooding. In October 2019, a very high flood rate occurred in Lagdo, while Obiafu and Kainji/Jebba experienced high and low flooding, respectively. In October 2018, the flood rankings were high for Obiafu, low for Kainji/Jebba, and very low for Lagdo.

4.4 September flooding analysis (Table 3)

In September 2022, Obiafu experienced very little flooding, Kainji/Jebba experienced low flooding, and Lagdo experienced very high flooding. In the previous year, September 2021, very low flooding occurred across all three areas. In September 2020, Lagdo experienced very little flooding, whereas Obiafu and Kainji/Jebba experienced very high flooding. September 2019 recorded high flooding in Obiafu, while Kainji/Jebba and Lagdo experienced low flooding. In September 2018, all three areas experienced very high flooding.

4.5 August flooding analysis (Table 4)

In August 2022, very low flood rankings were observed for Obiafu and Kainji/Jebba, while low flood rankings were observed for Lagdo. Similarly, in August 2021, the rankings were very low for Obiafu and Kainji/Jebba and low for Lagdo. The flood rankings were very low across all three locations in August 2020. In August 2019 and 2018, Obiafu and Kainji/Jebba maintained very low flood rankings, while Lagdo experienced high and very high flooding, respectively. The key observations were that the flooding intensity varied significantly across the study period. Obiafu experienced very high

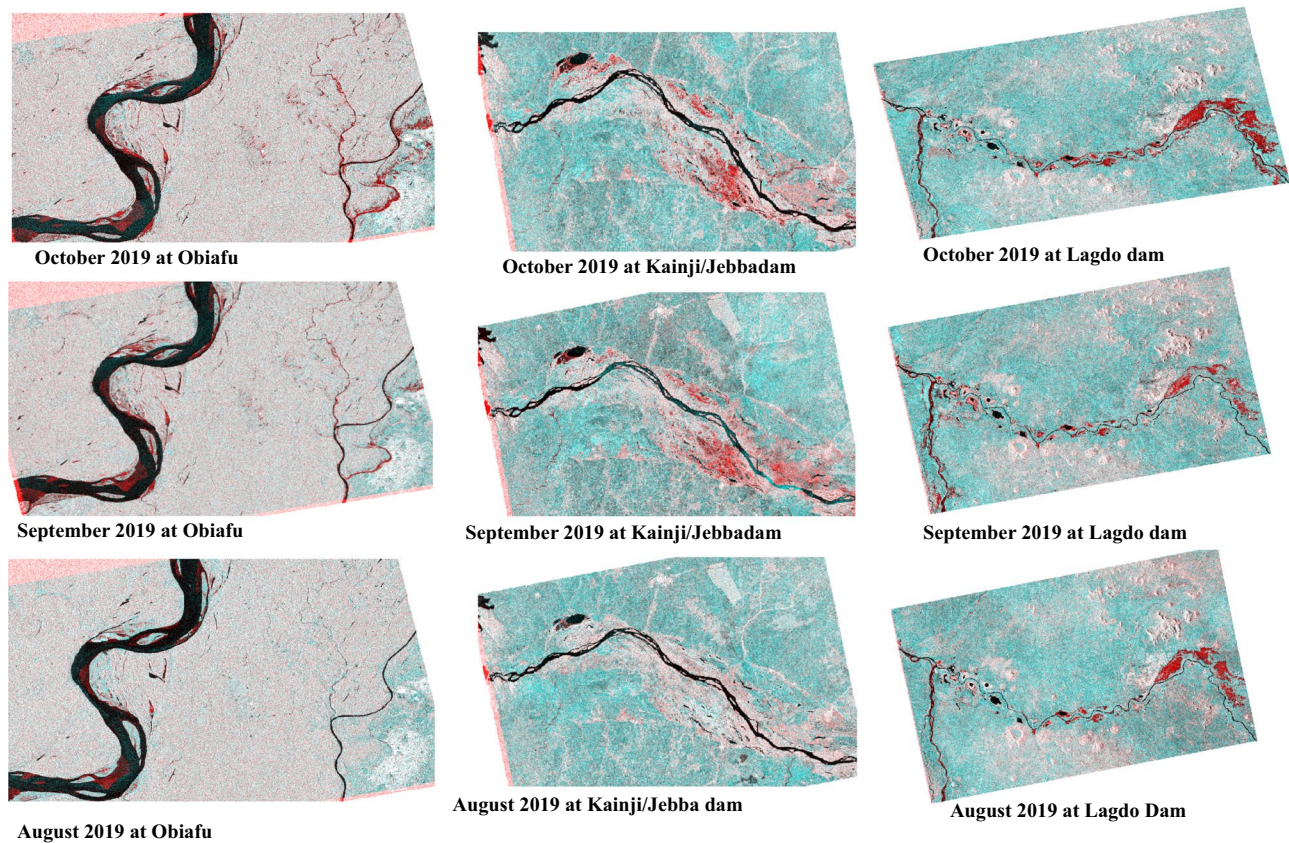


Fig. 6 SAR model result from August 2019 to October 2019

flooding in October 2020 and 2022, while lower flood levels generally occurred in September and August. This reveals the potential influence of seasonal rainfall patterns on flooding in Obiafu. The flood ranking for Obiafu sometimes differed from the Kainji/Jebba dam ranking, which showed that dam management practices at the Kainji/Jebba dam may not solely determine flooding in Obiafu. This could be due to factors such as the distance between the dam and the community or the influence of local tributaries on the flooding of Obiafu.

4.6 Comparison of water levels: upstream to downstream

Comparing actual water levels at hydrological stations from upstream (Jebba and Lagdo) to downstream (Obiafu) highlights the influence of reservoir operations on flood conditions. Hydrographs representing water levels at these stations from 2018 to 2022 (Figs. 8a–c) provide insight into the temporal relationship between upstream water release and downstream flooding.

The hydrograph simulation for Jebba (Fig. 8a) shows a consistent rise in water levels during the rainy season, particularly from July to September. During these months, water inflows from upstream reach their peak due to increased rainfall, necessitating controlled releases from the dam to prevent overtopping. The hydrograph for Lagdo (Fig. 8b) exhibits similar trends, with peak water levels occurring during the same period. This indicates a strong correlation between rainfall patterns and reservoir inflows, which in turn affect downstream water levels.

The downstream hydrograph for Obiafu (Fig. 8c) shows a delayed response compared to the upstream stations, with the peak water levels typically occurring 1 to 2 weeks after the maximum inflow at Jebba and Lagdo. This delay reflects the time required for water to travel downstream from the reservoirs, highlighting the importance of coordinated water release strategies between the dams to minimize downstream flood risks. The peak water levels observed at Obiafu coincide with periods of maximum water release from the upstream reservoirs, indicating that upstream reservoir operations significantly influence flood conditions in the Niger Delta.

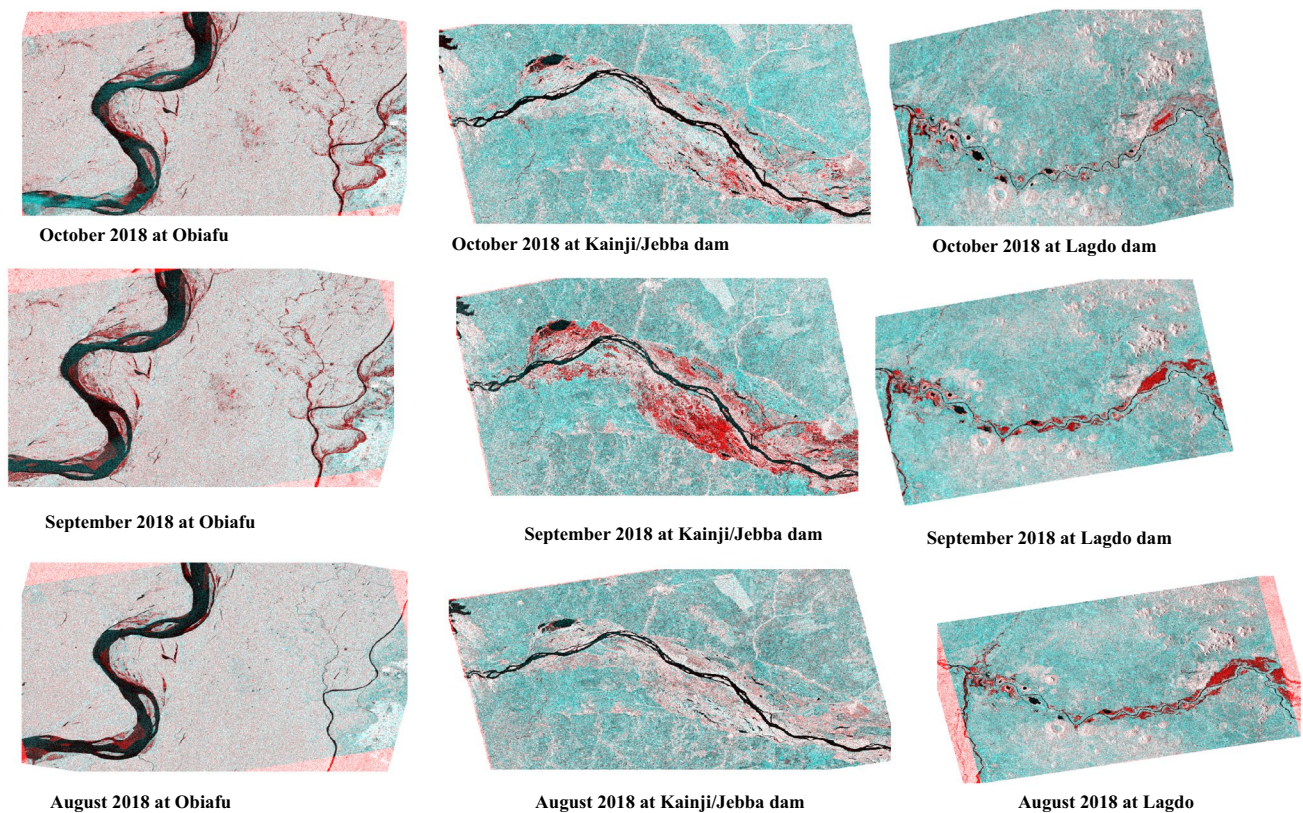


Fig. 7 SAR model result from August 2018 to October 2018

Table 1 Flood ranking level with reference to the SAR results

S/N	Flood Ranking level	Notation
1	Very low	Blue
2	Low	Green
3	High	Yellow
4	Very High	Red

4.7 Hydrograph simulation of water levels from 2018 to 2022

The hydrograph simulations for Jebba, Lagdo, and Obiafu water level variations over a 5-year period. These simulations are crucial for understanding the temporal patterns of flooding and the impact of upstream reservoir operations on downstream flood conditions. The Jebba hydrograph shows seasonal variations in water levels, with peak flows occurring during the rainy season (July to September) in Fig. 8a. The water levels at Jebba are directly influenced by rainfall patterns and reservoir operations, with controlled releases occurring during periods of high inflow to prevent flooding upstream. The hydrograph indicates a consistent rise in water levels during the rainy season, followed by a gradual decline as water is released downstream. The hydrograph for Lagdo shows similar trends to Jebba, with peak water levels occurring during the rainy season in Fig. 8b. The timing of water releases from Lagdo is coordinated with Jebba to minimize downstream flood risks. However, the hydrograph also shows occasional spikes in water levels due to extreme rainfall events, which can lead to rapid increases in downstream water levels. The hydrograph for Obiafu shows the downstream response to water releases from Jebba and Lagdo (Fig. 8c). The peak water levels at Obiafu typically occur one to 2 weeks after the peak inflows at Jebba and Lagdo, reflecting the time required for water to

Table 2 SAR model flood ranking for October

S/no	Year	Flood Ranking at Obiafu	Flood Ranking at Kainji/Jebba dam	Flood Ranking at Lagdo dam	Date Obiafu	Date Kainji/Jebba dam	Date Lagdo dam
1	2022	Very high	Very high	Very High	6/10/2022	6/10/2022	10/10/2022
2	2021	Very low	Very low	Very low	18/10/2021	11/10/2021	15/10/2021
3	2020	Very high	Very High	Low	16/10/2020	16/10/2020	8/10/2020
4	2019	High	Low	Very High	22/10/2019	10/10/2019	14/10/2019
5	2018	High	Low	Very low	22/10/2018	15/10/2018	7/10/2018

Table 3 SAR model flood ranking for September

S/no	Year	Flood Ranking at Obiafu	Flood Ranking at Kainji/Jebba dam	Flood Ranking at Lagdo dam	Date Obiafu	Date Kainji/Jebba dam	Date Lagdo dam
1	2022	Very low	High	Very High	12/9/2022	12/09/2022	16/09/2022
2	2021	Very low	Very low	Very low	24/09/2021	17/09/2021	9/9/2021
3	2020	Very high	Very High	Very low	22/09/2020	18/09/2020	14/09/2020
4	2019	High	Low	Low	28/09/2019	16/09/2019	20/09/2019
5	2018	Very High	Very High	Very High	28/09/2018	21/09/2018	13/09/2018

Table 4 SAR model flood ranking for August

S/no	Year	Flood Ranking Obiafu	Flood Ranking at Kainji/Jebba dam	Flood Ranking at Lagdo dam	Date Obiafu	Date Kainji/Jebba dam	Date Lagdo dam
1	2022	Very Low	Very Low	Low	26/08/2022	19/8/2022	23/8/2022
2	2021	Very Low	Very low	Low	19/08/2021	12/8/2021	16/8/2021
3	2020	Very Low	Very low	Very low	17/08/2020	15/08/2020	21/08/2020
4	2019	Very Low	Very low	High	23/08/2019	11/8/2019	27/8/2019
5	2018	Very Low	Very low	Very High	23/08/2018	16/08/2018	20/08/2018

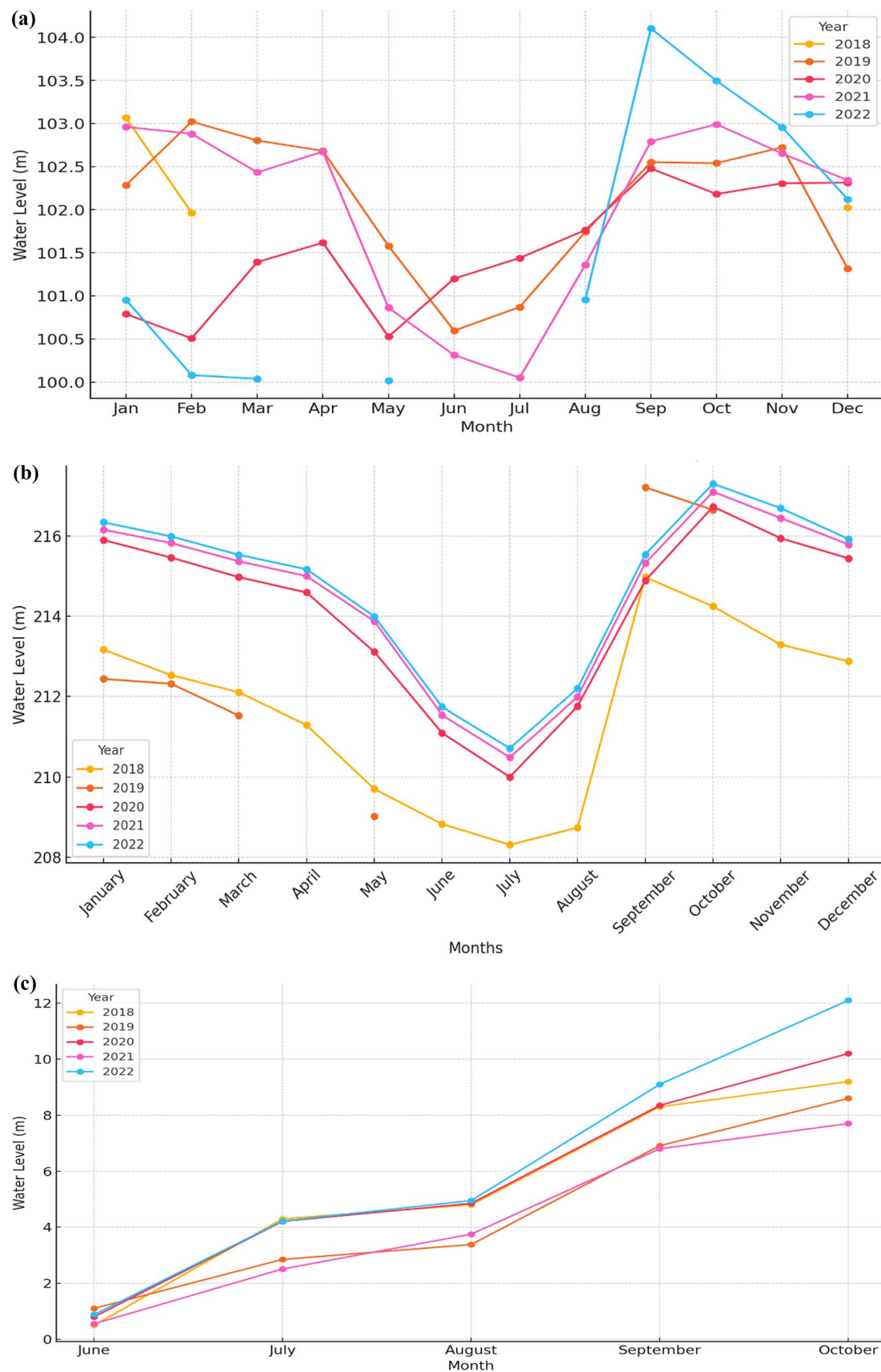


Fig. 8 **a** Hydrograph simulation of water level from 2018 to 2022 in station at Jebba (Upstream). **b** Hydrograph simulation of water level from 2018 to 2022 in station at Iagdo (upstream). **c** Hydrograph simulation of water level from 2018 to 2022 in station at Obiafu (Niger Delta- downstream)

travel downstream. The hydrograph shows significant flooding during the rainy season, with water levels remaining high for extended periods due to the slow drainage of floodwaters in the low-lying Niger Delta.

4.8 Alignment of Sentinel-1 SAR data with largest inundation depths

The analysis of Sentinel-1 SAR data for flood monitoring is highly dependent on the timing of the satellite pass relative to the peak flood events. In this study, the SAR data was collected during periods of maximum inundation depth to ensure accurate flood extent mapping. The largest inundation depths typically occurred in September and October, aligning with the peak water levels observed in the downstream hydrograph at Obiafu. The Sentinel-1 SAR data for these months was compared with actual water level measurements at the hydrological stations to ensure consistency between the modeled flood extents and observed water levels. The timing of the Sentinel-1 SAR imagery was carefully chosen to correspond with the periods of highest water levels in both upstream and downstream stations, ensuring that the model accurately captured the extent of flooding during peak flow periods.

4.9 SAR model flood rankings for August, September, and October

The SAR model flood rankings for August, September, and October provide a quantitative assessment of flood severity during these months. Tables 2, 3, and 4 with Tables 5, 6 and 7 summarize the flood rankings based on the extent of inundation observed in the SAR data for each month. The rankings are based on factors such as the area covered by floodwaters, the depth of inundation, and the duration of flooding. Flood severity is typically quantified through various parameters, such as the area inundated, the depth of floodwaters, and the duration of the flood event. The Sentinel-1 SAR flood model produced rankings for flood events in August, September, and October. Tables 5, 6, and 7 present the flood rankings based on SAR data for these months. The flood severity rankings for each month were derived from the extent of inundation, the depth of floodwaters, and the overall flood duration. The SAR data provided insights into the temporal dynamics of flooding in the Niger Delta, with peak flood events observed in September and October.

During August, moderate flooding was observed, particularly in areas adjacent to major rivers and floodplains. The SAR model indicated localized flooding in the upper and middle regions of the delta due to heavy rainfall events (Table 5 and Fig. 9). Although the area inundated was substantial, the flood depths and duration were relatively lower compared to subsequent months.

September saw the highest flood rankings, with extensive inundation across the Niger Delta (Table 6). The peak water releases from Jebba and Lagdo dams during this month led to widespread flooding downstream. The SAR data captured significant inundation across the region, affecting farmlands, settlements, and infrastructure. The depth and duration of flooding in September were higher compared to other months, marking this as the period of greatest flood risk.

Table 5 Inundated of SAR flood levels for August

Region	Area Inundated (km ²)	Average depth (m)	Flood duration (Days)	Ranking
Jebba	120	4.8	5	Moderate
Lagdo	95	4.5	3	Moderate
Obiafu	130	4.9	7	Moderate

Table 6 Inundated of SAR flood levels for September

Region	Area inundated (km ²)	Average depth (m)	Flood duration (Days)	Ranking
Jebba	180	5.5	12	Severe
Lagdo	150	5.7	10	Severe
Obiafu	210	6.0	14	Severe

Table 7 Inundated of sar flood levels for October

Region	Area inundated (km ²)	Average depth (m)	Flood duration (Days)	Ranking
Jebba	140	5.4	9	Severe
Lagdo	120	5.1	7	Moderate
Obiafu	180	5.8	10	Severe

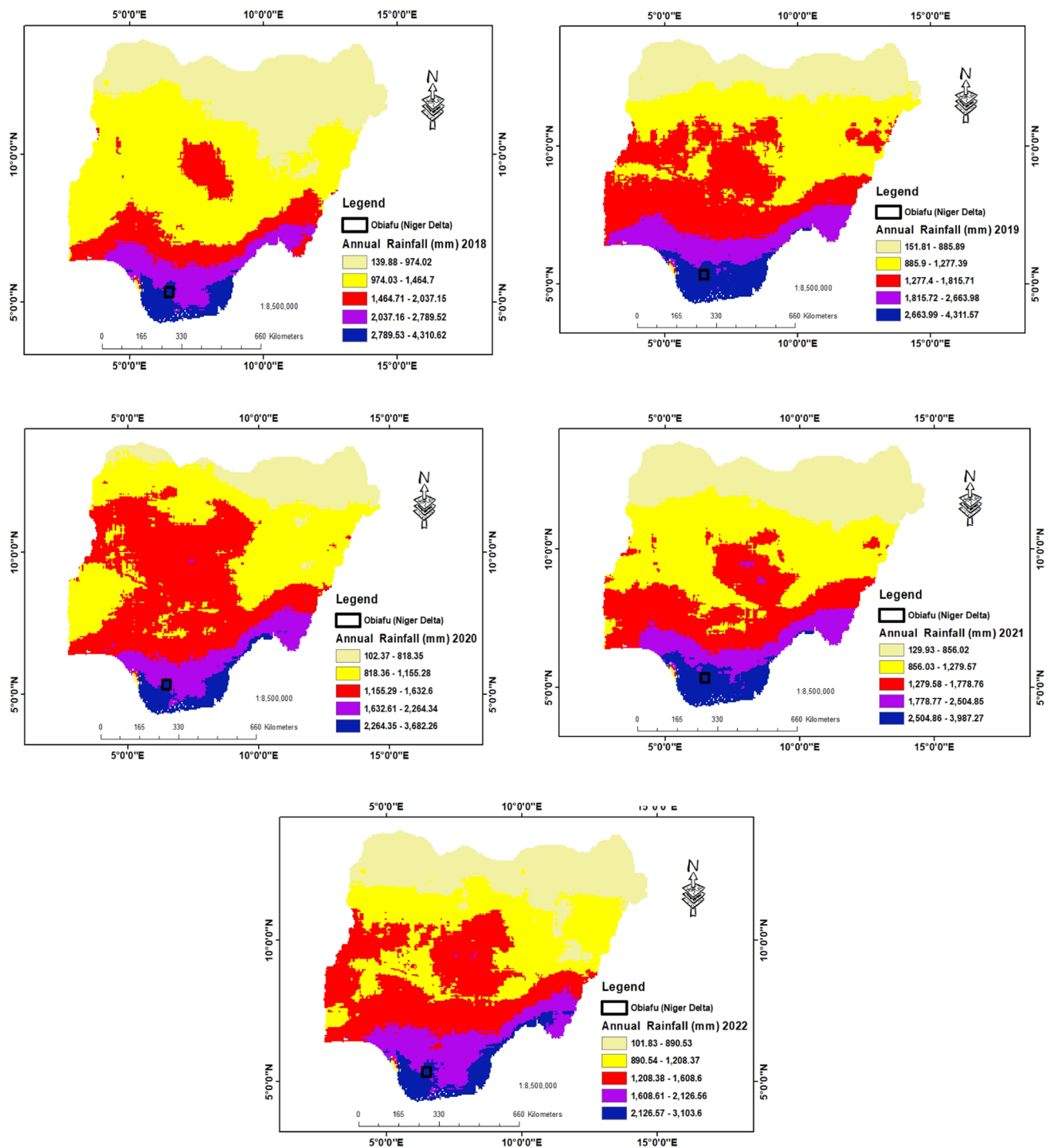


Fig. 9 Annual rainfall from 2018 to 2022 in Nigeria

In October, floodwaters began to recede as water levels in upstream reservoirs were reduced (Table 7). However, several regions, particularly in the lower Niger Delta, remained inundated due to the slow drainage of water in low-lying areas. The SAR data indicated a slight decrease in flood severity compared to September, though large areas remained underwater.

4.10 Rainfall patterns and flooding in the Niger Delta

Rainfall is a key driver of flooding in the Niger Delta, and its variability across different years significantly affects flood severity. The region experiences heavy rainfall during the wet season, with peak periods typically occurring between June and October. However, localized weather variations and broader climatic trends often lead to fluctuations in annual and monthly rainfall amounts.

From 2018 to 2022, rainfall patterns upstream near the Kainji/Jebba and Lagdo dams showed considerable fluctuations, as depicted in Fig. 9. In 2018, the upstream areas received annual rainfall ranging from 974.03 to 1464.7 mm, whereas the downstream region of Obiafu in the Niger Delta experienced significantly higher rainfall totals, with a range between 2789.53 and 4310.62. These values suggest a distinct difference in precipitation amounts between the upstream dam areas and the downstream deltaic region. In 2019, a slight decrease in rainfall was observed upstream, with values ranging from 885.9 to 1277.39, while downstream in the Niger Delta, the rainfall remained consistently high. The year 2020 saw further reductions in rainfall in both upstream and downstream regions, with upstream totals ranging between 818.36 and 1155.28 and downstream figures between 2264.35 and 3682.26. This disparity in rainfall distribution underlines the critical importance of understanding how both upstream and downstream rainfall patterns influence flood risks in the region. The Niger Delta receives higher rainfall primarily due to its proximity to the Atlantic Ocean, where moist winds bring significant precipitation during the rainy season. Conversely, the upstream dams (Kainji, Jebba, and Lagdo) receive relatively lower rainfall as they are situated farther inland in the savanna and semi-arid zones, which experience less rainfall intensity. This geographical difference creates a challenge for dam operators to balance water releases in response to upstream inflows and avoid exacerbating flood risks downstream during periods of high rainfall. The reduction in rainfall upstream could be attributed to broader climatic changes affecting the Sahel and West African Monsoon systems. This period also coincided with intensified dam operations, which compounded the flood risks downstream. By 2021, upstream rainfall totals had begun to increase again, reaching between 856.03 and 1279.57 mm, whereas downstream, rainfall continued to rise, with totals between 2504.86 and 3987.27 mm. The increase in precipitation upstream heightened flood risks downstream, especially when combined with dam releases. In 2022, rainfall patterns upstream and downstream remained high, with values ranging between 890.54 and 1208.37 mm in the Kainji/Jebba and Lagdo dam areas and between 2126.57 and 3103.6 mm in the Obiafu region. The rainfall distribution in the Niger Delta correlates closely with flood patterns in the region, especially during peak rainy seasons between July and October. The concentration of rainfall in these months often leads to the saturation of soils, overflow of rivers, and inundation of low-lying areas. Downstream areas like Obiafu in the Niger Delta are particularly susceptible to flooding due to their lower elevation and proximity to rivers, as shown in Fig. 1, which presents the Digital Elevation Model (DEM) of the study area. This DEM indicates that the Niger Delta lies significantly lower than the upstream areas, with elevation ranges from 78 to 35 m, compared to the higher elevations near Kainji and Jebba dams (235.11 m to 342.73 m). This topographical difference exacerbates the flood risk downstream, as water released from the dams converges towards the lower regions, overwhelming local drainage capacities.

4.11 Correlation between Sentinel-1 data and rainfall results for upstream and downstream Nigeria

Relationships between dam operation, rainfall patterns, and flooding in the Niger Delta region of Nigeria. The analysis focused on the Kainji/Jebba and Lagdo Dams located upstream of the Niger Delta. Tables 2, 3, 4 and Figs. 3, 4, 5, 6, 7 indicates a strong correlation between dam operations, particularly water releases, and flooding events in Obiafu town (Niger Delta). Rainfall data from Essen during the study period confirmed that heavy precipitation significantly contributed to flooding in the Niger Delta, causing both flash floods and river floods. However, the research suggests that rainfall, particularly water level rise, might not be the sole factor influencing flood severity. Rainfall data from 2021 reveal an inverse relationship with flooding in Obiafu, despite rainfall levels comparable to those in 2018, a year when the Niger Delta experienced a major flood (following similar events in 2012 and 2022). Interestingly, Sentinel-1 satellite imagery revealed flooding patterns near the Kainji/Jebba and Lagdo dams, suggesting that dam management practices, in addition to heavy rainfall, may be another contributing factor to flooding in the Niger Delta. The low-lying nature of the Niger Delta, coupled with the high runoff and infiltration rates evident in the Sentinel-1 data, amplifies the impact of uncontrolled water release from these dams. This uncontrolled release is likely a significant contributor to the seasonal flooding observed in the region.

4.12 Dam management and flooding in the Niger Delta

The management of dams such as Kainji, Jebba, and Lagdo plays a critical role in regulating water levels along the Niger and Benue rivers. These dams are essential for hydropower generation, irrigation, and water supply, but they also influence flood dynamics in downstream regions. Flooding occurs when the dams release excess water, especially during periods of peak rainfall. The correlation between dam releases and flood peaks in the Niger Delta is particularly evident during the months of September and October, as shown in Tables 2, 3. September marks the onset of heightened flood risk, with upstream dam releases contributing to riverine flooding in downstream areas. The SAR model flood ranking for September, illustrated in Table 3, shows significant inundation, particularly in low-lying areas of the Niger Delta, such as Obiafu. In October, flood risks further escalate, with extensive inundation reported downstream, as captured by the SAR model flood ranking for October (Table 2). In 2018 and 2020, for instance, high rainfall upstream combined with dam water releases led to severe flooding in the Niger Delta, affecting communities along the Orashi River and other tributaries. Figure 10 illustrates the water movement from the upstream River Benue and Niger River systems to the downstream Orashi River in the Niger Delta. The synchronization of dam releases with peak rainfall events in the downstream region creates a heightened flood risk, particularly during years of anomalous rainfall patterns. Figure 9 highlight how higher-than-average rainfall in the Niger Delta in 2021 coincided with increased dam releases, resulting in widespread flooding.

Water movement from upstream dams to the Niger Delta follows a clear pattern influenced by dam operations and rainfall. During periods of heavy rainfall, the Lagdo and Kainji/Jebba dams release large volumes of water downstream to prevent dam overtopping, which increases the risk of flooding in low-lying areas of the Delta.

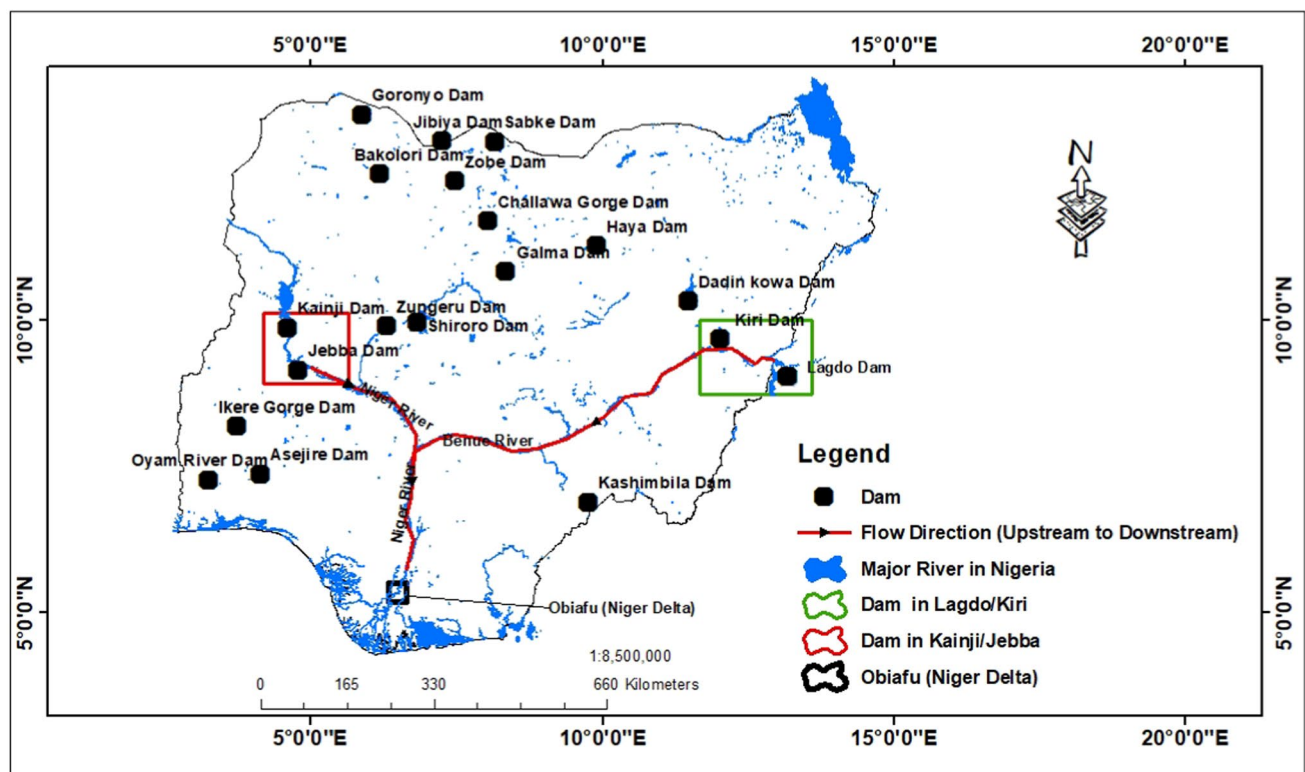


Fig. 10 Map Showing Water Movement from Upstream to Downstream Areas

5 Conclusion

This study highlights the significant influence of dam operations and rainfall patterns on flooding in the Niger Delta, utilizing Sentinel-1 SAR data for precise flood mapping and analysis. The results provide information on the temporal dynamics of flood events, revealing the complex interaction between rainfall, dam releases, and flood severity. The main findings of the study are as follows:

1. **Sentinel-1 SAR Effectiveness:** Sentinel-1 SAR data, combined with the RGB band technique, effectively identified flooded areas and distinguished them from permanent water bodies in Obiafu, capturing flood events in August, September, and October over 5 years.
2. **Flood Ranking Across Time:** Flood severity varied significantly across the study period, with the highest flood levels recorded in September 2022. The SAR model flood rankings revealed that both rainfall intensity and dam releases significantly influenced flooding patterns, particularly in Obiafu and downstream regions.
3. **Influence of Reservoir Operations:** The hydrograph simulations indicated that dam management at Kainji, Jebba, and Lagdo played a crucial role in flood dynamics, particularly during the rainy season. Water releases from upstream reservoirs were found to have a delayed but significant impact on downstream water levels, contributing to severe flooding in low-lying areas like the Niger Delta.
4. **Rainfall and Flood Correlation:** There was a strong correlation between rainfall patterns and flood severity. The Niger Delta experienced higher rainfall compared to upstream regions, which, combined with dam water releases, exacerbated flood risks during peak rainy months.
5. **Dam Management and Flood Risk:** The study demonstrates a strong correlation between dam management practices and downstream flood risk. Uncontrolled or poorly coordinated water releases during peak rainfall periods significantly contribute to the severity of flooding in the Niger Delta.
6. **Geographical Influence on Flooding:** The Niger Delta's low elevation and proximity to major rivers exacerbate flood risks, particularly during periods of high rainfall and upstream dam releases, which overwhelm local drainage systems.
7. **Socio-Economic Impacts:** The recurrent flooding in the Niger Delta has severe socio-economic impacts, disrupting agriculture, fishing, transportation, and infrastructure, highlighting the need for better flood management practices.
8. **Integration of SAR Data for Flood Mitigation:** The study emphasizes the importance of integrating SAR data with hydrological models and early warning systems to improve flood prediction accuracy and enable timely disaster response in vulnerable regions.

Therefore, the combination of Sentinel-1 SAR data with rainfall and dam operation records offers a powerful tool for flood monitoring and risk mitigation in the Niger Delta. Improved coordination between upstream dam management and flood response strategies is essential to reduce the adverse effects of seasonal flooding in the region.

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Author contributions E.D.R., E.E.F and, O.A developed the study with part of the writing and Formatting, E.D.R., I.O, O.A and P.M processed the data, and E.E.F, I.O, P, M and E.D. R assisted with review. The final version was co-written by all the authors.

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Data availability This study explored how dam operation and rainfall patterns interact to influence flooding in the Niger Delta, Nigeria. The following data supporting the findings of this study are publicly available: 1. Satellite Data: Sentinel-1 satellite data (European Space Agency (ESA), 2018–2022) were used to analyze flood extents. These data can be accessed through the ESA's Copernicus Open Access Hub (<https://scihub.copernicus.eu/>). 2. Rainfall Data: Rainfall data were obtained from the Center for Hydrometeorology and Remote Sensing (CHRS) for the period 2018–2022. These data are available at the CHRS's data portal (<http://chrsdata.eng.uci.edu/>). 3. The Shuttle Radar Topography Mission (SRTM) data were obtained through satellite imagery provided by the United States Geological Survey (USGS) <https://earthexplorer.usgs.gov>. 4. Surface water level data for Kainji/Jebba and Lagdo dams were collected using the Database for Hydrological Time Series of Inland Waters (DAHITI) from the Deutsches Geodätisches Forschungsinstitut at Technische Universität München <https://dahiti.dgfi.tum.de/en/1472/surfa-ce-area/>. This system employs satellite altimetry to monitor inland water levels, ensuring accurate and continuous data. Complementing this, data from the Nigeria Hydrological Services Agency (NIHSA) were integrated to provide a comprehensive hydrological time series. The combined datasets allow for better understanding of water level fluctuations and enhance flood monitoring and management in the region. Rainfall Data: Rainfall data for the period 2018–2022 was obtained from the Center for Hydrometeorology and Remote Sensing (CHRS). You can access this data through the CHRS's data portal <http://www.chrs.web.uci.edu/>.

Declarations

Competing interests The authors declare no competing interests.

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